1 Introduction to Chemistry

Learning Outcomes

The children will be able to:

- discuss the importance of Chemistry in daily life and its role in different industries and life processes;
- list important applications of Chemistry in day to day life;
- list some industrial applications of Chemistry;
- discuss the bio-sketches of some great scientists and their works;
- appreciate the patience, perseverance, sacrifices and ethical conduct of scientists.

UNIT 1 - CHEMISTRY AND ITS IMPORTANCE

SCIENCE OF CHEMISTRY

3WCEDAOJ2F

The branch of science which deals with the study of materials (substances), especially about their composition, methods of preparation, properties and their reactions with other substances, is called **chemistry**.

From the materials we mean common substances around us, such as air, water, stones, various kinds of plants and animals. It includes natural compounds such as rocksalt, marble, petroleum; **metals** like gold, silver, copper, etc.; **animal products** like milk, cheese, butter, etc.; **vegetable products** like wheat, maize, oil seeds, etc In materials, we can include **man-mad products** like cement, glass, steel, plastic, bronze, nylon, etc.

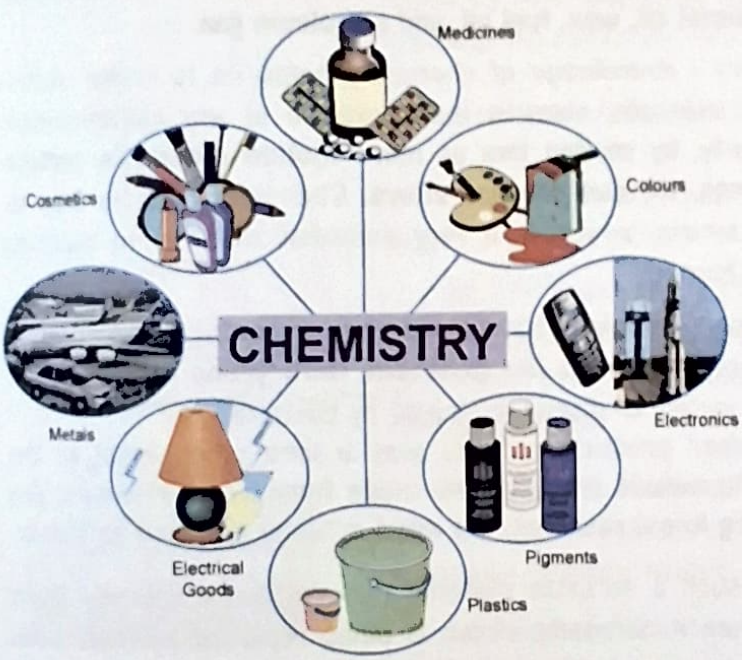


Fig. 1.1

In the study of a particular material, the chemists perform experiments with that material under various conditions. They make very careful observations and finally make conclusions regarding the behaviour of the material.

In this process, sometimes chemists discover other useful materials, which go a long way in making human life comfortable.

IMPORTANCE OF CHEMISTRY

Chemistry is of such vital importance that practically there is no human activity which is directly or indirectly not dependent on it.

For example, the book you are reading is made possible only because chemists found methods for the production of paper from wood. They developed ink which does not spread on paper. They developed metals from which printing machine is made. Following are the broad fields in which knowledge of chemistry is employed extensively.

(a) **Agriculture** : The food supply to any society entirely depends on agriculture. However, with the increase in population, the pressure on land has increased for the production of more food.

Chemistry has helped agriculture in the following ways :

(i) It helped in the production of artificial fertilisers which increase food production.

(ii) It helped in the production of better seeds.

(iii) It helped in producing chemicals which kill insects (**insecticides**), fungi (fungicides) and unwanted herbs (**herbicides**).

DID YOU KNOW ?

The petroleum gas obtained during the refining of petroleum is cooled and compressed under very high pressure when it gets liquefied and is commonly called liquefied petroleum gas (LPG). It is filled in steel cylinders and is marketed as household fuel.

(b) **Mineral prospecting** : Man needs a variety of metals for making various kinds of machines and structures. Chemistry helps us in the extraction of metals from their ores.

Similarly, petroleum is mined from underground reservoirs. Chemistry shows the way of obtaining useful products from petroleum, such as **petrol, kerosene oil, diesel oil, wax, fuel oil, and petroleum gas**.

(c) **Industry** : Knowledge of chemistry helps us to make useful materials. For example, steel is the backbone of any industrialised country. Similarly, **by mixing two or more molten metals in certain fixed proportions, we can prepare alloys.** Chemistry tells us how to manufacture cement, which is a very essential material in building industries and homes.

SPOTLIGHT

The alloys have far superior properties than metals. For example, duralumin is an alloy of aluminium, copper, magnesium and manganese which is very light, yet as tough as steel. It is used for making aircraft frames. Similarly, we can have alloys which resist corrosion, such as stainless steel.

(d) **Medicine** : Chemistry has helped us in discovering new drugs to fight against diseases. Drugs like **penicillin, tetracycline** and **ampicillin** fight against a variety of diseases caused by bacteria.

(e) **Consumer products** : There was a time when most of the furniture and household articles were made from wood. However, due to fast dwindling forest reserves, the wood is being replaced by plastic.

Plastic is such a versatile material that we can practically make many things from it. Similarly, cotton is being replaced by man-made fibres such as **rayon, nylon** and **terylene**.

DID YOU KNOW ?

Common salt, sugar, vinegar and citric acid are common household preservatives. Similarly, sodium metabisulphite and sodium benzoate are powerful artificial food preservatives.

Thus, chemistry is helping in the conservation of forest wealth, which is very essential for balance of the ecosystem. Furthermore, it is providing cheap and durable substitutes for us.

(f) Household : Chemistry has provided food preservatives which prevent wastage of food. Thus, the saved food is used for feeding millions of hungry people. In kitchen, brass and aluminium utensils have been replaced by stainless steel. The liquefied petroleum gas (LPG) is used for cooking.

(g) Cosmetic industry : All kinds of powders, creams, nail polishes, lipsticks, cleansing lotions, soaps, detergents, etc., are the result of research done by the chemists. These researches help us to live in a clean environment.

DID YOU KNOW ?

Good laboratory apparatus is made from a special kind of glass, called pyrex glass or borosil glass. This glass expands very little on heating and does not crack. Furthermore, it is resistant to chemicals and is transparent to see what happens during a chemical reaction.

(h) Protecting materials from corrosion : Paints, varnishes, greases, etc., protect a large number of articles of daily use from the effects of moist air which causes corrosion.

Furthermore, chemistry has helped us to develop artificial dyes, which are widely used in making paints and for printing purposes.

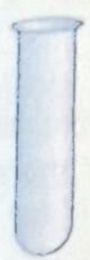
(i) Electronics : Research in chemistry has provided versatile materials such as **silicon** and **germanium**, which are widely used in making diodes. It is the diode which makes electrical circuits in radios, televisions, computers, calculators, etc.

(j) Space research : The present day space technology is the direct result of production of better and more powerful fuels. It is the discovery of these fuels which has helped us to launch space satellites.

Thus, to sum up, we can say that chemistry helps the mankind to shape its own destiny by modifying materials available in nature, or by synthesising materials, which are not available in nature.

Some common chemistry laboratory apparatus.

Some common chemistry laboratory equipments are shown in Figs. 1.2 (a) to 1.2 (t).

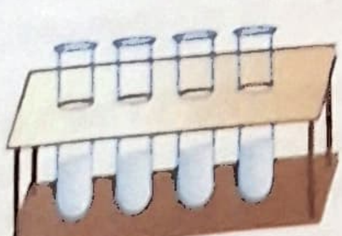


(a) Test tube

(a) Test tube : Fig. 1.2 (a). Test tubes of various sizes are made from different types of glass. A test tube which is heated directly on flame is made from pyrex glass and is commonly called **hard glass test tube** or **boiling test tube**.

(b) Test tube holder

(b) Test tube holder : Fig. 1.2 (b). It is a kind of an iron tong provided with wooden or plastic handle. It is used for holding a test tube when a substance is being heated.



(c) Test tube stand or

(c) Test tube stand or Test tube rack : Fig. 1.2 (c). It is a plastic or wooden stand for keeping test tubes in vertical position.

(d) Round bottomed flask : **Fig. 1.2 (d).** It is a glass container with a spherical bulb and a narrow cylindrical neck. It is generally used for heating liquids.

(e) Flat bottomed flask : **Fig. 1.2 (e).** It is a glass container with a spherical bulb, which is flattened at the base and is provided with cylindrical neck. It is used for the mixing/storing liquid chemicals.

bottomed
sk

(f) Conical flask : **Fig. 1.2 (f).** It is a cone shaped flask with a flat base and provided with a cylindrical neck. It is also used for mixing/ storing liquid chemicals.

(g) Beaker : **Fig. 1.2 (g).** It is an open glass container, cylindrical in shape and provided with a lip for pouring out liquids. The beakers are of different sizes such as 50 cc, 100 cc, 250 cc, 500 cc.

flask

(h) Glass tubing : **Fig. 1.2 (h).** It is a hollow glass tube of 3 mm diameter and open at both ends. It is generally used for shaping delivery tubes of various shapes by heating.

(i) Glass rod : **Fig. 1.2 (i).** It is a solid glass tube of 3 mm diameter. It is generally used for stirring liquid chemicals.

tubing

(j) Funnel : **Fig. 1.2 (j).** It is a conical vessel provided with long tapering neck and is made from glass or plastic. It is employed for pouring out liquids from one vessel to another without causing any spilling.

(k) China dish or Evaporating dish : **Fig. 1.2 (k).** It is made from porcelain. It is used for evaporating chemicals in the form of solutions by heating.

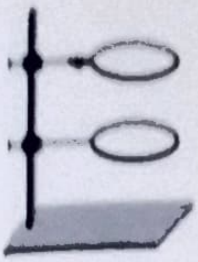
Funnel

(l) Pipette : **Fig. 1.2 (l).** It is a long narrow tube provided with a nozzle at one end and a bulb in the middle. A circular mark is made in its neck which signifies the volume of liquid which it can measure. It is used for measuring a fixed volume of liquid chemicals and then transferring it to another vessel.

(m) Burette : **Fig. 1.2 (m).** It is a long graduated tube provided with a glass stopcock at its bottom end. It has a capacity of 50 mL. It is used for pouring out a fixed volume of liquid (less than 50 ml) chemicals.

) Burette

(n) Measuring cylinder or Graduated cylinder : **Fig. 1.2 (n).** It is a cylindrical glass vessel provided with a flat base and a lip near the top. It is used for measuring a definite volume of a liquid and then pouring it out into another vessel.

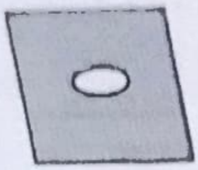


(o) Iron stand

(p) Tripod stand

(o) Iron stand : Fig. 1.2 (o). It is used for holding a glass apparatus (generally round bottomed flask or hard glass test tube) in a specific position.

(p) Tripod stand : Fig. 1.2 (p). It is a triangular hollow frame provided with three legs, and is used for supporting glass apparatus, which needs heating.



(q) Asbestos wire gauze

(q) Asbestos wire gauze : Fig. 1.2 (q). It is an iron wire mesh provided with a thin sheet of asbestos in the middle. It distributes heat from the burner evenly to the glass apparatus and hence prevents its cracking.

(r) Mortar and pestle

(r) Mortar and pestle : Fig. 1.2 (r). It is made from glazed porcelain. The solid substances are placed in mortar and then gently hammered with pestle so as to powder them.

(s) Spirit lamp : Fig. 1.2 (s). It is a flat cylindrical vessel made of glass or brass and provided with a brass neck through which a thick cotton wick passes. It is filled with methylated spirit. On ignition the spirit burns to produce very hot flame, which is used for heating chemicals.

(s) Spirit lamp

(t) Bunsen burner

(t) Bunsen burner : Fig. 1.2 (t). Modern chemistry laboratories use Bunsen burner, in place of spirit lamp, for heating purposes.

When the Bunsen burner is connected to the gas supply, the gas issues out of the nozzle with a good pressure and in doing so, sucks in the air from the air holes. This mixture of gas and air when ignited burns on the top of burner tube with blue flame.

Fig. 1.2 : (a) to (t)

Exercise 1.1

A. Fill in the blank spaces by choosing the correct words from the list given below :

List : metals, fertilisers, cement, reservoirs, pestle, hard, pipette, round, rod

1. Artificial fertilisers have helped farmers to grow more food.
2. Petroleum is mined from underground reservoirs
3. Cement is a very essential material for building industry.
4. Paints protect metals from corrosion.
5. A boiling tube is also called a hard glass test tube.
6. Solid substances are ground with the help of a pastle in a mortar.
7. A pipette is used to measure a fixed quantity of a liquid.
8. A round bottomed flask is used for heating liquids.
9. A glass rod is used for stirring liquids.